

Question:
Find the sum of all integer bases $b > 9$ for which 17_b is a divisor of 97_b .

Reasoning + Answer:

<think>

1. **Understand the problem**: We need to find the sum of all integer bases $(b > 9)$ for which (17_b) is a divisor of (97_b) .

2. **Convert the bases to decimal**:

- (17_b) in base (b) is $(1 \cdot b^1 + 7 \cdot b^0 = b + 7)$.
- (97_b) in base (b) is $(9 \cdot b^1 + 7 \cdot b^0 = 9b + 7)$.

3. **Set up the divisor condition**:

- $(b + 7)$ (the decimal value of (17_b)) must divide $(9b + 7)$ (the decimal value of (97_b)).

4. **Express the condition mathematically**:

- The condition $(b + 7)$ divides $(9b + 7)$ can be written as $(9b + 7 = k(b + 7))$ for some integer (k) .

5. **Solve the equation for (k)** :

- Rearrange the equation: $(9b + 7 = k(b + 7))$.
- Distribute (k) on the right-hand side: $(9b + 7 = kb + 7k)$.
- Subtract (kb) from both sides: $((9 - k)b + 7 = 7k)$.

6. **Isolate the terms involving (b)** :

- $((9 - k)b = 7k - 7)$.
- Divide both sides by $(9 - k)$: $(b = \frac{7k - 7}{9 - k})$.

7. **Determine the valid (b) values**:

- Since (b) must be an integer and greater than 9, we solve for (k) such that $(b > 9)$.

8. **Find integer solutions for (k)** :

- $(b = \frac{7k - 7}{9 - k})$.
- Test possible integer values of (k) to find valid (b) values.

9. **List the valid (b) values**:

- By testing values, we find $(b = 16)$, $(b = 23)$, and $(b = 31)$.

10. **Sum the valid (b) values**:

- $(b = 16 + 23 + 31 = 70)$.

</think>

<answer>

70

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